Def. Let (X,d) be a Metric space.
A sequence is said to be Convergent if, for some $x \in X$, for any $\varepsilon > 0$, there exist $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow d(x_n, x) < \varepsilon.$$

Lemma. Let $\{x_n\}, \{y_n\}$ be Convergent sequences in $\mathbb{R}$.

1. $\left(\lim_{n\to\infty} x_n\right) + \left(\lim_{n\to\infty} y_n\right) = \lim_{n\to\infty} (x_n + y_n)$

2. $\alpha \cdot \left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} (\alpha x_n)$.

3. $\left(\lim_{n\to\infty} x_n\right) \cdot \left(\lim_{n\to\infty} y_n\right) = \lim_{n\to\infty} (x_n \cdot y_n)$

4. If $\lim_{n\to\infty} y_n \neq 0$ and $y_i \neq 0 \ \forall i \in \mathbb{N}$, then $\left(\lim_{n\to\infty} y_n\right)^{-1} = \lim_{n\to\infty} y_n^{-1}$

Proof. Denote $x, y \in X$ are limit of $\{x_n\}, \{y_n\}$, respectively.
Let $\varepsilon > 0$ be given. Then, there exist $N_1, N_2 \in \mathbb{N}$ s.t

$$n \geq N_1 \Rightarrow |x_n - x| < \frac{\varepsilon}{2} \quad \text{and} \quad n \geq N_2 \Rightarrow |y_n - y| < \frac{\varepsilon}{2}$$

By the triangle inequality,

$$n \geq \max\{N_1, N_2\} \Rightarrow |x_n + y_n - (x_n + y_n)| \leq |x_n - x| + |y_n - y| < \varepsilon.$$

Let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow |x_n - x| < \frac{\varepsilon}{|\alpha|}$$

$$\Rightarrow |\alpha||x_n - x| = |\alpha x_n - \alpha x| < \varepsilon. \quad (\text{If } \alpha = 0, \text{ trivial. Thus let } \alpha \neq 0)$$

Let $\varepsilon > 0$ be given. Then, there exist $N_1, N_2 \in \mathbb{N}$ s.t

$$n \geq N_1 \Rightarrow |x_n - x| < \sqrt{\varepsilon} \quad \text{and} \quad n \geq N_2 \Rightarrow |y_n - y| < \sqrt{\varepsilon}.$$

Then, for $N = \max\{N_1, N_2\}$,

$$n \geq N \Rightarrow |(x_n - x)(y_n - y)| < \varepsilon$$

Therefore, $\lim_{n\to\infty} (x_n - x)(y_n - y) = 0$. Now, for all $n \in \mathbb{N}$,

$$0 \leq |x_n y_n - xy| = |x_n - x| \cdot |y_n - y| + x|y_n - y| + y|x_n - y| \to 0 \quad \text{as } n \to \infty,$$

Thus $x_n y_n \to xy$.
Let $\varepsilon > 0$ be given. Choose $N_1 \in \mathbb{N}$ s.t $n \geq N_1 \Rightarrow |y_n - y| < \frac{1}{2}|y|$. Then, $n \geq N_1 \Rightarrow \frac{|y|}{2} < |y_n|$
Meanwhile, choose $N_2 \in \mathbb{N}$ s.t $n \geq N_2 \Rightarrow |y_n - y| < \frac{1}{2}|y|^2 \varepsilon$. Now, for $n \geq \max\{N_1, N_2\}$,

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \left|\frac{y_n - y}{y_n y}\right| < \frac{2}{2} \cdot \frac{|y|^2}{|y|^2} \cdot \varepsilon = \varepsilon.$$

$\boxtimes$

Thm. Every Convergent sequence in a Metric space is Cauchy sequence.

Proof. Let $\{x_n\}$ be a Convergent sequence in a Metric space (X,d), which converges to $x \in X$.

Let $\varepsilon > 0$ be given. Chose $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow d(x_n, x) < \frac{\varepsilon}{2} \qquad d(a_m, a_n) \leq d(a_m, a_N) + d(a_n, a_N)$$

Then, by the triangle inequality,

$$m, n \geq N \Rightarrow d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon.$$

Thm. Every Cauchy sequence in a Compact Metric space is Convergent.

Proof. Let $\{a_n\}$ be a Cauchy sequence in a Compact Metric space (X, d).

Consider a map diameter of subset in Metric space

$$\operatorname{diam} E : \mathcal{P}(X) \to \mathbb{R} ; E \mapsto \sup\{ d(p,q) \mid p, q \in E \}$$

This map satisfies: $\operatorname{diam} E = \operatorname{diam} \overline{E}$, for all $E \subseteq X$, because: $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$ is clear, let $\varepsilon > 0$ be given. Then, for any $p', q' \in \overline{E}$, there exists $p, q \in E$ s.t

$$d(p, p') < \frac{\varepsilon}{2} \quad \text{and} \quad d(q, q') < \frac{\varepsilon}{2}$$

Now, the triangle inequality gives

$$d(p', q') \leq d(p', p) + d(p, q) + d(q, q') < \varepsilon + \operatorname{diam} E.$$

Therefore $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$, $\operatorname{diam} \overline{E} = \operatorname{diam} E$.

Take $E_N = \{ a_i \mid i \geq N \}$. By definition of Cauchy sequence, $\lim_{N \to \infty} \operatorname{diam} E_N = 0$.

Moreover, Combining above result,

$$\lim_{N \to \infty} \operatorname{diam} \overline{E_N} = \lim_{N \to \infty} \operatorname{diam} E_N = 0.$$

Since Closed set in Compact Space is Compact, and by definition, $\{\overline{E_n}\}$ is Compact sets s.t

$$\overline{E_1} \supseteq \overline{E_2} \supseteq \cdots \qquad \overline{E_i} \neq \phi.$$

Then $\bigcap_{n=1}^{\infty} \overline{E_n}$ is a singleton, because: if $\bigcap_{n=1}^{\infty} \overline{E_n}$ is empty, then

$$\overline{E_1} \cap \bigcap_{n=2}^{\infty} \overline{E_n} = \phi \text{ which implies } \overline{E_1} \subseteq \bigcup_{n=2}^{\infty} \overline{E_n}^c$$

Since $\overline{E_1}$ is compact, for some finite covers, $2 \leq n_1 < \cdots < n_r$,

$$\overline{E_1} \subseteq \overline{E_{n_1}}^c \cup \cdots \cup \overline{E_{n_r}}^c \text{ which implies } \overline{E_1} \cap \overline{E_{n_1}} \cap \cdots \cap \overline{E_{n_r}} = \overline{E_{n_r}} = \phi.$$

Contradiction. And, if it contains more than one point, then $\lim_{N \to \infty} \operatorname{diam} \overline{E_N} > 0$, Contradiction.

Finally, let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \operatorname{diam} \overline{E_n} < \varepsilon.$$

Now, for $a \in \bigcap_{n=1}^{\infty} \overline{E_n}$,

$$k \geq N \Rightarrow d(a_k, a) \leq \operatorname{diam} \overline{E_k} < \varepsilon.$$

Corollary. $\mathbb{R}^n$ is Complete Metric Space.

Proof. Every Cauchy sequence in a Metric Space is bounded, we can cover this by Compact subset of $\mathbb{R}^n$.

Special Sequence in $\mathbb{R}$.

Thm.

1) If $p>0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.
2) If $p>0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$
3). $\lim_{n\to\infty} \sqrt[n]{n} = 1$
4) If $\alpha\in\mathbb{R}$ and $x>1$, then $\lim_{n\to\infty} \frac{n^\alpha}{x^n} = 0$.

Proof. Let $\varepsilon>0$ be given.

$x = 1+p$

1) Choose $N = \left(\frac{1}{\varepsilon}\right)^{\frac{1}{p}}$. Then,

$$n \geq N \iff n \geq \left(\frac{1}{\varepsilon}\right)^{\frac{1}{p}} \Rightarrow \frac{1}{n^p} < \varepsilon$$

2) If $p=1$, trivial. If $p>1$, take $x_n = \sqrt[n]{p} - 1$. The Binomial Theorem gives

$$p = (x_n+1)^n = \sum_{k=0}^{n} \binom{n}{k} x_n^k \geq 1 + n \cdot x_n \text{ which implies } x_n \leq \frac{p-1}{n} \to 0 \text{ as } n\to\infty.$$

If $p<1$, $\lim_{n\to\infty} \sqrt[n]{p} = \lim_{n\to\infty} \frac{1}{\sqrt[n]{\frac{1}{p}}} = \frac{1}{\lim_{n\to\infty} \sqrt[n]{\frac{1}{p}}}$ converges.

3). Take $x_n = \sqrt[n]{n} - 1$. The Binomial Thm gives

$$n = (x_n+1)^n = \sum_{k=0}^{n} \binom{n}{k} x_n^k \geq \frac{n(n-1)}{2} x_n^2 \text{ which implies } x_n \leq \sqrt{\frac{2}{n-1}} \to 0 \text{ as } n\to\infty.$$

4). Take $k\in\mathbb{N}$ s.t. $k \geq \alpha$. Then, for all $n \geq 2k$,

$$x^n = (1+(x-1))^n \geq \binom{n}{k}(x-1)^k = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}\cdot(x-1)^k \geq \frac{n^k}{2^k}\cdot\frac{(x-1)^k}{k!}$$

which implies that

$$\frac{n^\alpha}{x^n} \leq n^{\alpha-k}\cdot\frac{2^k k!}{(x-1)^k} \to 0 \text{ as } n\to\infty.$$